

Harmonized survey questions for monitoring household energy use and SDG indicators 7.1.1 and 7.1.2

Concise roster format



This document contains a set of **questions essential to monitor SDG indicator 7.1.2 on primary reliance on clean fuels and technologies, as well as SDG 7.1.1 on the proportion of the population with access to electricity**. These questions were developed by the World Health Organization (WHO) and the World Bank Group in consultation with a diverse group of stakeholders convened in a set of several expert consultations, and have also been extensively piloted.

The essential questions for monitoring SDG 7.1.2 and SDG 7.1.1 are marked with an asterisk (*), totalling 13 questions. These questions are available in three versions to facilitate integration to existing surveys, including versions that align with common national surveys like the USAID's Demographic and Health Surveys (DHS) or UNICEF's Multiple Indicator Cluster Surveys (MICS). All three versions of survey questions capture the same key data:

1. fuels and technologies used the most time for cooking, heating and lighting (i.e. primary reliance);
2. all other fuels and technologies used by the household for cooking, heating and lighting;
3. electricity access in the household;
4. time use, fuel collection, injuries and health;
5. main source of electricity and appliances used (for solar energy);
6. availability of electricity supply;
7. reliability of electricity supply.

Supporting documentation for the use of the survey questions includes:

- *Household Energy Use Survey Question Guide*, designed for statistical agencies and programmes using the questions to understand the importance of each survey question, and how to calculate relevant indicators.
- *Household Energy Use Catalog*, includes detailed descriptions of all answer options for the survey questions, including pictorial examples.
- *Household Energy Use Survey Questions: Interviewer Manual*, designed for use in enumerator training, and as a reference tool for survey teams both preceding and during data collection.

If these household energy questions are implemented following the instructions included in the supporting documentation, consistent and comparable monitoring of household energy use patterns across time and populations will be possible.

Updated November 2019

Overview of general instructions for administering questions

- Questions should be administered to a reference person in the household (a person who knows about cooking, heating and lighting practices, such as the head of the household or main cook).
- Questions are provided that guide the interviewer to skip certain sections if cooking, heating or lighting are not performed in the household. Additionally, certain countries may choose to omit the module on heating if it is not applicable.
- Questions can be adapted to both pen and paper or electronic data collection methodologies.
- Questions are written to be administered through one-on-one, in-person interviews, which can be part of national census and household surveys or suitable for use in smaller projects.
- Italicized text indicates instructions to the interviewer. This text should not be read aloud to the respondent.
- Interviewers should not read response options to respondents after administering a question, but should select the response which best fits the information given by the respondent. If the information given is not enough to select a fitting response(s), the interviewer should neutrally ask for more information by asking, for example, “Could you please give me a bit more information?”
- Manufactured and traditional solid fuel cookstove definitions (more details and photos will be provided in the supporting documentation mentioned above):
 - *Manufactured solid fuel cookstoves*: The category for manufactured cookstoves includes any cookstove that is produced and sold or distributed to the end user (including “traditional” cookstoves that are manufactured). For some manufactured cookstoves, specialized technicians come to the household to install/build the cookstove. Manufactured cookstoves are often constructed from prefabricated materials or materials usually not readily available in the immediate community. Manufactured cookstoves are often designed with improvements or advanced design features such as an improved and enclosed combustion chamber, high quality materials, and/or high quality production processes – unlike traditional solid fuel cookstoves.
 - *Traditional solid fuel cookstoves (non-manufactured)*: Traditional cookstoves are usually constructed at the home. Traditional cookstoves are made with local materials, such as mud, ceramics or clay. These cookstoves do not generally contain advanced design features but may include limited improvements over a three-stone cookstove or open fire (e.g. a pot rest, simple enclosed combustion spaces).

Direction for interviewer: Please inform the respondent that you will now ask questions about cooking, fuel collection, heating, lighting, and electricity use.

HOUSEHOLD COOKING	
HC1. Is any food or drink consumed by household members cooked or prepared at the household dwelling using a cookstove, fire or other cooking device?	Yes 1 No 2 → <i>Skip to Heating module</i>
HC2. How many stoves (including open fires) are used for these activities?	Number of cookstoves..... If the respondent reports fewer than 3 cookstoves, leave extra columns (below) blank. If respondent reports use of more than 3 cookstoves, use additional sheets.

	a. 1st Cookstove (MAIN)	b. 2nd Cookstove	c. 3rd Cookstove
HC3. * What does this household use for cooking most of the time, including cooking food, making tea/coffee, boiling drinking water? Please tell me the cookstove or device that is used for the most time, followed by the other cookstove(s) or device(s) used most often, if applicable. (Phrase question based on response to HC2. Select one type for each cookstove.)	Solar cooker (thermal energy, not solar panels) 01 → HC7 Electric stove..... 02 → HC6 Piped natural gas stove 03 → HC6 Biogas stove..... 04 → HC6 Liquefied petroleum gas (LPG)/cooking gas stove... 05 → HC6 Manufactured solid fuel stove 06 Traditional solid fuel stove (non-manufactured)..... 07 Liquid fuel stove..... 08 → HC5 Moveable firepan 09 → HC5 Three stone stove/ open fire 10 → HC5 Other, specify 96	Solar cooker (thermal energy, not solar panels) 01 → HC7 Electric stove..... 02 → HC6 Piped natural gas stove 03 → HC6 Biogas stove..... 04 → HC6 Liquefied petroleum gas (LPG)/cooking gas stove... 05 → HC6 Manufactured solid fuel stove 06 Traditional solid fuel stove (non-manufactured)..... 07 Liquid fuel stove..... 08 → HC5 Moveable firepan 09 → HC5 Three stone stove/ open fire 10 → HC5 Other, specify 96	Solar cooker (thermal energy, not solar panels) 01 → HC7 Electric stove..... 02 → HC6 Piped natural gas stove 03 → HC6 Biogas stove..... 04 → HC6 Liquefied petroleum gas (LPG)/cooking gas stove... 05 → HC6 Manufactured solid fuel stove 06 Traditional solid fuel stove (non-manufactured)..... 07 Liquid fuel stove..... 08 → HC5 Moveable firepan 09 → HC5 Three stone stove/ open fire 10 → HC5 Other, specify 96
HC4. * What is the brand of the cookstove or device?	(Pre-fill with name or brand of manufactured solid fuel stoves available in country – may add additional lines.) NAME/BRAND 1 01 NAME/BRAND 2..... 02 NAME/BRAND 3..... 03 Other, specify 96	(Pre-fill with name or brand of manufactured solid fuel stoves available in country – may add additional lines.) NAME/BRAND 1 01 NAME/BRAND 2..... 02 NAME/BRAND 3..... 03 Other, specify 96	(Pre-fill with name or brand of manufactured solid fuel stoves available in country – may add additional lines.) NAME/BRAND 1 01 NAME/BRAND 2..... 02 NAME/BRAND 3..... 03 Other, specify 96
HC5. * What type of fuel or energy source does this household use most of the time in this cookstove or device for cooking food, making tea/coffee and boiling drinking water?	Alcohol/ethanol 01 Gasoline/diesel (not in generator) 02 Kerosene/paraffin 03 Coal/lignite unprocessed 04 Coal/lignite briquettes/ pellets..... 05 Charcoal unprocessed..... 06 Charcoal briquettes/pellets . 07 Wood 08 Agricultural or crop residue/ grass/straw/shrubs/ corn cobs..... 09 Animal waste/dung..... 10 Processed biomass pellets/ briquettes 11 Woodchips 12 Garbage/plastic 13 Sawdust..... 14 Other 96	Alcohol/ethanol 01 Gasoline/diesel (not in generator) 02 Kerosene/paraffin 03 Coal/lignite unprocessed 04 Coal/lignite briquettes/ pellets..... 05 Charcoal unprocessed..... 06 Charcoal briquettes/pellets . 07 Wood 08 Agricultural or crop residue/ grass/straw/shrubs/ corn cobs..... 09 Animal waste/dung..... 10 Processed biomass pellets/ briquettes 11 Woodchips 12 Garbage/plastic 13 Sawdust..... 14 Other 96	Alcohol/ethanol 01 Gasoline/diesel (not in generator) 02 Kerosene/paraffin 03 Coal/lignite unprocessed 04 Coal/lignite briquettes/ pellets..... 05 Charcoal unprocessed..... 06 Charcoal briquettes/pellets . 07 Wood 08 Agricultural or crop residue/ grass/straw/shrubs/ corn cobs..... 09 Animal waste/dung..... 10 Processed biomass pellets/ briquettes 11 Woodchips 12 Garbage/plastic 13 Sawdust..... 14 Other 96
HC6. How much did this household pay for this fuel or energy source last month for cooking (the last 30 days)? (in local currency)	_____ (local currency) Pays nothing 01 Does not know 99	_____ (local currency) Pays nothing 01 Does not know 99	_____ (local currency) Pays nothing 01 Does not know 99
HC7. In the past 12 months , how often was this fuel or energy source unavailable in the quantity you desired?	Often (more than once a month)..... 01 Sometimes (4–12 times a year) 02 Rarely (less than 4 times a year) 03 Never (always available) 04 Does not know / unsure 99	Often (more than once a month)..... 01 Sometimes (4–12 times a year) 02 Rarely (less than 4 times a year) 03 Never (always available) 04 Does not know / unsure 99	Often (more than once a month)..... 01 Sometimes (4–12 times a year) 02 Rarely (less than 4 times a year) 03 Never (always available) 04 Does not know / unsure 99

HOUSEHOLD COOKING (continued)			
	a. 1 st cookstove (MAIN)	b. 2 nd cookstove	c. 3 rd cookstove
HC8. What other fuels and energy sources does this household use in this cookstove or device for cooking food, making tea/coffee, boiling drinking water and/or starting the fire? <i>(Circle all that respondent mentions.)</i>	No others..... 01 Alcohol/ethanol 02 Gasoline/diesel (not in generator) 03 Kerosene/paraffin 04 Coal/lignite unprocessed 05 Coal/lignite briquettes/ pellets..... 06 Charcoal unprocessed..... 07 Charcoal briquettes/ pellets..... 08 Wood 09 Agricultural or crop residue/grass/straw/ shrubs/corn cobs 10 Animal waste/dung..... 11 Processed biomass pellets/briquettes..... 12 Woodchips..... 13 Garbage/plastic..... 14 Sawdust..... 15 Other 96	No others..... 01 Alcohol/ethanol 02 Gasoline/diesel (not in generator) 03 Kerosene/paraffin 04 Coal/lignite unprocessed 05 Coal/lignite briquettes/ pellets..... 06 Charcoal unprocessed..... 07 Charcoal briquettes/ pellets..... 08 Wood 09 Agricultural or crop esidue/grass/straw/ shrubs/corn cobs 10 Animal waste/dung..... 11 Processed biomass pellets/briquettes..... 12 Woodchips..... 13 Garbage/plastic..... 14 Sawdust..... 15 Other 96	No others..... 01 Alcohol/ethanol 02 Gasoline/diesel (not in generator) 03 Kerosene/paraffin 04 Coal/lignite unprocessed 05 Coal/lignite briquettes/ pellets..... 06 Charcoal unprocessed..... 07 Charcoal briquettes/ pellets..... 08 Wood 09 Agricultural or crop esidue/grass/straw/ shrubs/corn cobs 10 Animal waste/dung..... 11 Processed biomass pellets/briquettes..... 12 Woodchips..... 13 Garbage/plastic..... 14 Sawdust..... 15 Other 96
HC9. Yesterday, how much time was this cookstove used for cooking food, making tea/coffee, and boiling drinking water?	Number of hours..... __ __ Number of minutes..... __ __ Does not know / unsure 99	Number of hours..... __ __ Number of minutes..... __ __ Does not know / unsure 99	Number of hours..... __ __ Number of minutes..... __ __ Does not know / unsure 99
HC10. How often did you use the cookstove or cooking device over the last week (last 7 days) for these activities?	Several times each day..... 01 About once per day..... 02 A few times this week..... 03 About once this week 04 Less than once this week ... 05 Does not know 99	Several times each day..... 01 About once per day..... 02 A few times this week..... 03 About once this week 04 Less than once this week ... 05 Does not know 99	Several times each day..... 01 About once per day..... 02 A few times this week..... 03 About once this week 04 Less than once this week ... 05 Does not know 99
HC11. Is the cooking usually done in the house, in a separate building, or outdoors? <i>(If in main house, probe to determine if cooking is done in a separate room. If outdoors, probe to determine if cooking is done on veranda, covered porch, or open air.)</i>	In main house: no separate room..... 01 In main house: separate room..... 02 Outside of main house: in a separate room 03 Outside of main house in open air 04→HC13 On veranda or covered porch..... 05→HC13 Other 96	In main house: no separate room..... 01 In main house: separate room..... 02 Outside of main house: in a separate room 03 Outside of main house in open air 04→HC13 On veranda or covered porch..... 05→HC13 Other 96	In main house: no separate room..... 01 In main house: separate room..... 02 Outside of main house: in a separate room 03 Outside of main house in open air 04→HC13 On veranda or covered porch..... 05→HC13 Other 96
HC12. Does the cookstove have a chimney or hood?	Yes 01 No 02 Don't know..... 99	Yes 01 No 02 Don't know..... 99	Yes 01 No 02 Don't know..... 99
HC13. In the past 12 months, did any harm or injury happen from using this cookstove, device or fuel? <i>(Circle all that respondent mentions.)</i>	None 01 Person burned 02 Fire in house..... 03 Poisoning..... 04 Death..... 05 Other 96	None 01 Person burned 02 Fire in house..... 03 Poisoning..... 04 Death..... 05 Other 96	None 01 Person burned 02 Fire in house..... 03 Poisoning..... 04 Death..... 05 Other 96

HOUSEHOLD ENERGY AND GENDER			
HG1. Enumerator/CAPI check: Is the cookstove used most of the time (HC3) electric, solar or piped natural gas?		Yes 01 →HG6 No 02	
HG2. Who usually goes to collect the main fuel for the cookstove your household uses most of the time ? <i>Record the name of the person who spends the most time collecting the main fuel HC5 for cookstove (a) and copy the line number of this person from the LIST OF HOUSEHOLD MEMBERS module. If multiple people spend the same time collecting, add additional names and line numbers.</i>		Members do not collect..... 00 → HG6 Name Line number <i>(If multiple collect for the same amount of time.)</i> Second person name Second person line number	
HG3. On a single trip, how long does it take for this person to go to collect the fuel, get the fuel, and come back?		Number of hours..... Number of minutes..... Does not know / unsure 99	
HG4. In the past month (the last 30 days), how many times has this person collected this fuel for household cooking?		Number of times in past month..... Does not know / unsure..... 99	
HG5. In the past 12 months, did this person experience an injury while collecting or transporting fuel? <i>(Prompt: Circle all.)</i>		No injury..... 00 Back pain 01 Back, neck or shoulder injury..... 02 Cuts or scrapes 03 Snake or animal bite 04 Other 05 Does not know / unsure..... 99	
HG6. Yesterday, how much time in total was spent preparing the [COOKSTOVE] and fuel for cooking, including setting up the fuel and lighting/turning on the cookstove but not including gathering fuel or cooking time? <i>(Time spent by anyone, not use person in HG2.)</i>		Number of hours..... Number of minutes..... Does not know / unsure 99	
HG7. Who in the household does most of the cooking, including cooking food, making tea/coffee and boiling drinking water? <i>Record the name of the person and copy the line number of this person from the List of Household Members module.</i>		Main cook is not household member 00 Name..... Line number	
HG8. Yesterday, how much time did this person spent cooking, including cooking food, tea/coffee, and boiling drinking water for household consumption?		Number of hours..... Number of minutes..... Does not know / unsure 99	
HOUSEHOLD HEATING			
HH1. Does this household use any heating device or fire to keep the dwelling/living quarters warm at any time during the year?		Yes 1 No 2 → Skip to Lighting module	
HH2. In the last 12 months, during how many months did you use a heating device or fire to keep the dwelling/living quarters warm?		Number of months.....	
HH3. How many space heaters, or heating devices (including cookstoves and open fires), are used in or around the dwelling to produce heat at any time during the year?		Number of heaters..... <i>If the respondent reports using fewer than 3 heaters to produce heat in the household, leave extra columns blank. If respondent reports using more than 3 heaters, use additional sheets.</i>	
	a. 1 st heater (MAIN)	b. 2 nd heater	c. 3 rd heater
HH4. * What does this household use to heat the home when needed? For example, do you use a space heater(s), or use your cookstove in the winter for warmth? Please tell me the heaters, cookstoves or heat systems used for the most time , [followed by the other heaters, cookstoves or devices used most often, if applicable]. <i>(Phrase question based on response to HH2, and select one type for each heater.)</i>	Central heating..... 01→HH6 Manufactured space heater 02 Traditional space heater 03 Manufactured cookstove ... 04 Traditional cookstove (non-manufactured)..... 05 Moveable heating pan 06→HH6 Open fire/three-stone stove..... 07→HH6 Heat pump 08→HH6 Other, specify..... 96	Central heating..... 01→HH6 Manufactured space heater 02 Traditional space heater 03 Manufactured cookstove ... 04 Traditional cookstove (non-manufactured)..... 05 Moveable heating pan 06→HH6 Open fire/three-stone stove..... 07→HH6 Heat pump 08→HH6 Other, specify..... 96	Central heating..... 01→HH6 Manufactured space heater 02 Traditional space heater 03 Manufactured cookstove ... 04 Traditional cookstove (non-manufactured)..... 05 Moveable heating pan 06→HH6 Open fire/three-stone stove..... 07→HH6 Heat pump 08→HH6 Other, specify..... 96
HH5. * What is the brand of the heater, cookstove or device?	<i>(Pre-fill with name or brand of devices available in country – may add additional lines.)</i> NAME/BRAND 1 01 NAME/BRAND 2..... 02 NAME/BRAND 3..... 03 Other, specify..... 96	<i>(Pre-fill with name or brand of devices available in country – may add additional lines.)</i> NAME/BRAND 1 01 NAME/BRAND 2..... 02 NAME/BRAND 3..... 03 Other, specify..... 96	<i>(Pre-fill with name or brand of devices available in country – may add additional lines.)</i> NAME/BRAND 1 01 NAME/BRAND 2..... 02 NAME/BRAND 3..... 03 Other, specify..... 96

HOUSEHOLD HEATING (continued)			
	a. 1 st heater (MAIN)	b. 2 nd heater	c. 3 rd heater
HH6. * What type of fuel or energy source does this household use most of the time for heating in this heater, cookstove or device?	Electricity (including solar panels) 01→HH9 Piped natural gas 02→HH9 LPG/cooking gas 03→HH9 Biogas 04→HH9 Alcohol/ethanol 05→HH8 Gasoline/diesel (not in generator) 06→HH8 Kerosene/paraffin 07→HH8 Coal/lignite unprocessed 08 Coal/lignite briquettes/ pellets 09 Charcoal unprocessed 10 Charcoal briquettes/ pellets 11 Wood 12 Agricultural or crop residue/grass/ straw/ shrubs/corn cobs 13 Animal waste/dung 14 Processed biomass pellets/briquettes 15 Woodchips 16 Garbage/plastic 17 Sawdust 18 Other 96	Electricity (including solar panels) 01→HH9 Piped natural gas 02→HH9 LPG/cooking gas 03→HH9 Biogas 04→HH9 Alcohol/ethanol 05→HH8 Gasoline/diesel (not in generator) 06→HH8 Kerosene/paraffin 07→HH8 Coal/lignite unprocessed 08 Coal/lignite briquettes/ pellets 09 Charcoal unprocessed 10 Charcoal briquettes/ pellets 11 Wood 12 Agricultural or crop residue/grass/ straw/ shrubs/corn cobs 13 Animal waste/dung 14 Processed biomass pellets/briquettes 15 Woodchips 16 Garbage/plastic 17 Sawdust 18 Other 96	Electricity (including solar panels) 01→HH9 Piped natural gas 02→HH9 LPG/cooking gas 03→HH9 Biogas 04→HH9 Alcohol/ethanol 05→HH8 Gasoline/diesel (not in generator) 06→HH8 Kerosene/paraffin 07→HH8 Coal/lignite unprocessed 08 Coal/lignite briquettes/ pellets 09 Charcoal unprocessed 10 Charcoal briquettes/ pellets 11 Wood 12 Agricultural or crop residue/grass/ straw/ shrubs/corn cobs 13 Animal waste/dung 14 Processed biomass pellets/briquettes 15 Woodchips 16 Garbage/plastic 17 Sawdust 18 Other 96
HH7. Does it have a chimney or hood?	Yes 01 No 02 Does not know 99	Yes 01 No 02 Does not know 99	Yes 01 No 02 Does not know 99
HH8. Which other fuels or energy sources are used for heating in this space heater, cookstove, device or fuel? (Circle all that respondent mentions.)	None 01 Alcohol/ethanol 02 Gasoline/diesel (not in generator) 03 Kerosene/paraffin 04 Coal/lignite unprocessed 05 Coal/lignite briquettes/ pellets 06 Charcoal unprocessed 07 Charcoal briquettes/ pellets 08 Wood 09 Agricultural or crop residue/grass/straw/ shrubs/corn cobs 10 Animal waste/dung 11 Processed biomass pellets/briquettes 12 Woodchips 13 Garbage/plastic 14 Sawdust 15 Other 96	None 01 Alcohol/ethanol 02 Gasoline/diesel (not in generator) 03 Kerosene/paraffin 04 Coal/lignite unprocessed 05 Coal/lignite briquettes/ pellets 06 Charcoal unprocessed 07 Charcoal briquettes/ pellets 08 Wood 09 Agricultural or crop residue/grass/straw/ shrubs/corn cobs 10 Animal waste/dung 11 Processed biomass pellets/briquettes 12 Woodchips 13 Garbage/plastic 14 Sawdust 15 Other 96	None 01 Alcohol/ethanol 02 Gasoline/diesel (not in generator) 03 Kerosene/paraffin 04 Coal/lignite unprocessed 05 Coal/lignite briquettes/ pellets 06 Charcoal unprocessed 07 Charcoal briquettes/ pellets 08 Wood 09 Agricultural or crop residue/grass/straw/ shrubs/corn cobs 10 Animal waste/dung 11 Processed biomass pellets/briquettes 12 Woodchips 13 Garbage/plastic 14 Sawdust 15 Other 96
HH9. In the past 12 months, did any harm or injury happen from using this space heater, cookstove or device? (Circle all that respondent mentions.)	None 01 Person burned 02 Fire in house 03 Poisoning 04 Death 05 Other 96	None 01 Person burned 02 Fire in house 03 Poisoning 04 Death 05 Other 96	None 01 Person burned 02 Fire in house 03 Poisoning 04 Death 05 Other 96

HOUSEHOLD LIGHTING

HL1. Does this household use anything for lighting?	Yes 1 No 2 → <i>Skip to Electricity module</i>
HL2. How many different sources of light are used in the dwelling/ living quarters?	Number of sources of light <i>If the respondent reports using fewer than 3 light sources in the household, leave extra columns in the following table blank. If respondent reports using more than 3 light sources, use additional sheets.</i>

	a. 1st light source (MAIN)	b. 2nd light source	c. 3rd light source
HL3. *What does this household use most of the time as energy for lighting, or as a light source? Please tell me the light source used for the most time each day, followed by other light sources used. (Select one type for each light source.)	Electricity (including solar panels) 01 Solar-powered lantern or flashlight 02 Rechargeable flashlight, mobile, torch or lantern 03 Battery powered flashlight, torch or lantern 04 Biogas lamp 05 LPG lamp 06 Gasoline lamp 07 Kerosene/paraffin lamp 08 Oil lamp 09 Candle 10 Open fire 11 Other 96	Electricity (including solar panels) 01 Solar-powered lantern or flashlight 02 Rechargeable flashlight, mobile, torch or lantern 03 Battery powered flashlight, torch or lantern 04 Biogas lamp 05 LPG lamp 06 Gasoline lamp 07 Kerosene/paraffin lamp 08 Oil lamp 09 Candle 10 Open fire 11 Other 96	Electricity (including solar panels) 01 Solar-powered lantern or flashlight 02 Rechargeable flashlight, mobile, torch or lantern 03 Battery powered flashlight, torch or lantern 04 Biogas lamp 05 LPG lamp 06 Gasoline lamp 07 Kerosene/paraffin lamp 08 Oil lamp 09 Candle 10 Open fire 11 Other 96
HL4. In the past 12 months, did any harm or injury happen from using this lighting source? (Circle all that respondent mentions.)	None 01 Person burned 02 Fire in house 03 Poisoning 04 Death 05 Other 96	None 01 Person burned 02 Fire in house 03 Poisoning 04 Death 05 Other 96	None 01 Person burned 02 Fire in house 03 Poisoning 04 Death 05 Other 96

HOUSEHOLD ELECTRICITY

HE1. * What source of electricity is used most of the time in this household? (Please circle one.)	No electricity in household 00 → Skip to next module <i>(Please customize options for each country.)</i> National grid connection from (Company)..... 01 → HE4 Local mini grid 02 → HE4 Solar home system..... 03 Solar lantern 04 Electric generator..... 05 Rechargeable battery 06 Dry cell battery/torch 07 → HE4 Other, specify..... 96 → HE4
HE2. * What appliances are powered using this household's solar device/system? (Please circle all that apply.)	Mobile phone charger 01 Radio..... 02 Television 03 Fan..... 04 Refrigerator..... 05 Electric iron..... 06 Cooking device..... 07 Other, specify..... 96
HE3. How many lightbulbs can be powered using this household's solar device/system?	Zero..... 01 One 02 Two or more 03 After this question, skip to HE5
HE4. Who does this household currently pay for [name main electricity system]?	<i>(Please customize options for each country.)</i> No one..... 00 Energy company/national utility 01 Pre-paid meter card seller 02 Community/village/municipality 03 Relative..... 04 Neighbour..... 05 Landlord 06 Local store 07 Utility office 08 Bank 09 Post office..... 10 Other..... 96
HE5. Are there any other sources of electricity used in this household? (Please circle all that apply.)	No other sources..... 00 National grid connection 01 Local mini grid 02 Solar home system 03 Solar lantern 04 Electric generator..... 05 Rechargeable battery 06 Dry cell battery / torch 07 Other 96
HE6. * In the last 7 days, how many hours of electricity were available each day on average from [name main electricity system]? (Maximum 24 hours.)	Number of hours..... — — Number of minutes — — Does not know / unsure 99
HE7. * In the last 7 days, how many hours of electricity were available each evening on average, from 6:00pm to 10:00pm from [name main electricity system]? (Maximum 4 hours.)	Number of hours..... — — Number of minutes — — Does not know / unsure 99
HE8. * In the last 7 days, how many times were there unscheduled outages or blackouts from [name main electricity system]?	Number of outages or blackouts..... — — Don't know / unsure 99
HE9. * What is the total duration of all the unscheduled outages or blackouts in the last 7 days?	Number of hours..... — — Number of minutes — — Don't know / unsure 99
HE10. Enumerator/CAPI check: Is the household's main source of electricity a national grid or local mini grid? <i>If HE1 is 1 or 2, mark "Yes"</i>	Yes 1 No 2 → Skip to next module
HE11. In the last 12 months, did any of this household's appliances get damaged because the voltage was going up and down in the [name main electricity system from HE1]?	Yes 1 No 2
HE12. In the last 12 months, did anyone using [name main electricity system from HE1] die or have permanent limb (bodily injury) damage?	Yes 1 No 2